

Series and Integral Transforms

Theorems & Formulas Reference Sheet

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Contents

1	Numerical Series	3
1.1	Basic Definitions and Facts	3
1.2	Convergence Tests	3
2	Power Series	4
2.1	Radius of Convergence	4
2.2	Taylor and Maclaurin Series	4
2.3	Product and Quotient of Power Series	4
2.4	Common Maclaurin Series	5
2.5	Taylor's Theorem with Remainder	6
2.6	Applications: Limits and Integrals via Taylor	6
3	Uniform Convergence	6
3.1	Sequences of Functions	6
3.2	Properties Preserved by Uniform Convergence	6
3.3	Series of Functions	7
4	Inner Product Spaces & L^2 Norm	7
4.1	Inner Product on $L^2[-L, L]$	7
4.2	Orthogonality of the Trigonometric Basis	8
4.3	Bessel's Inequality and Parseval's Identity	8
5	Fourier Series	8
5.1	Definition and Coefficients	8
5.2	Pointwise Convergence: Dirichlet's Theorem	8
5.3	Even, Odd, and Half-Period Extensions	9
5.4	Shift Property	9
5.5	Parseval's Identity and Computing Numerical Series	9
5.6	Complex Fourier Series	10
5.7	Summing a Real Fourier Series in Closed Form	10
5.8	Hyperbolic and Complex Functions (for Fourier)	11
5.9	Technique: Complex Fourier of e^{ax} on $[-L, L]$	11

6	Fourier Transform	12
6.1	Definition and Inverse (ω convention)	12
6.2	Key Properties	12
6.3	Common Transform Pairs	13
6.4	Parseval / Plancherel Theorem	13
7	Laplace Transform	13
7.1	Definition	13
7.2	Common Transforms and Properties	13

1 Numerical Series

1.1 Basic Definitions and Facts

- The series $\sum_{n=1}^{\infty} a_n$ **converges** if the partial sums $S_N = \sum_{n=1}^N a_n$ have a finite limit.
- **Necessary condition:** if $\sum a_n$ converges, then $a_n \rightarrow 0$. Contrapositive: if $a_n \not\rightarrow 0$, the series diverges.
- **Absolutely convergent:** $\sum |a_n| < \infty$ (implies convergent).
- **Conditionally convergent:** $\sum a_n$ converges but $\sum |a_n|$ diverges.
- **p-series:** $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges $\Leftrightarrow p > 1$.

1.2 Convergence Tests

Test	Preconditions	Conclusion
Comparison	$a_n, b_n \geq 0$ (positive series)	If $a_n \leq b_n$ and $\sum b_n < \infty$, then $\sum a_n < \infty$. If $a_n \geq b_n \geq 0$ and $\sum b_n = \infty$, then $\sum a_n = \infty$.
Limit Comparison	$a_n, b_n > 0$ (positive series)	If $\lim \frac{a_n}{b_n} = L \in (0, \infty)$, then $\sum a_n$ and $\sum b_n$ behave the same.
Ratio (D'Alembert)	$a_n \neq 0$; tests <i>absolute</i> convergence	$L = \lim \left \frac{a_{n+1}}{a_n} \right $: converges if $L < 1$, diverges if $L > 1$, inconclusive if $L = 1$.
Root (Cauchy)	Any series; tests <i>absolute</i> convergence	$L = \limsup a_n ^{1/n}$: converges if $L < 1$, diverges if $L > 1$.
Integral	f continuous, positive, <i>decreasing</i> on $[1, \infty)$; $a_n = f(n)$ (conditions need hold only eventually)	$\sum_{n=1}^{\infty} f(n)$ and $\int_1^{\infty} f(x) dx$ converge or diverge together.
Alternating (Leibniz)	$b_n \geq 0$; $b_n \searrow 0$ monotonically	$\sum (-1)^n b_n$ converges.
Dirichlet	Partial sums of $\sum a_n$ bounded; $b_n \searrow 0$ monotonically	$\sum a_n b_n$ converges.
Abel	$\sum a_n$ converges; b_n monotone and bounded	$\sum a_n b_n$ converges.

Remark (Parity trick: $\cos(n^k \pi) = (-1)^n$ for odd k). For any integer n and *odd* positive integer k : $n^k \equiv n \pmod{2}$, so $\cos(n^k \pi) = (-1)^{n^k} = (-1)^n$. Similarly $\sin(n^k \pi) = 0$ for

any positive integer k . Use this to simplify series whose terms contain $\cos(n^3\pi)$, $\cos(n^5\pi)$, etc.

Theorem 1.1 (Alternating Series Error Bound). *If $\sum (-1)^n b_n$ satisfies the Leibniz conditions ($b_n \geq 0$, $b_n \searrow 0$), then the error after N terms satisfies $|S - S_N| \leq b_{N+1}$. To achieve accuracy $< \varepsilon$: use the smallest N with $b_{N+1} < \varepsilon$.*

Theorem 1.2 (Telescoping Series). *If $a_n \rightarrow L$ (finite), then $\sum_{n=1}^{\infty} (a_n - a_{n+1}) = a_1 - L$, since $S_N = a_1 - a_{N+1} \rightarrow a_1 - L$. Recognition: a series written as $f(n) - f(n+1)$ (or $f(n+1) - f(n)$) is telescoping.*

2 Power Series

2.1 Radius of Convergence

The series $\sum_{n=0}^{\infty} c_n (x - a)^n$ converges absolutely for $|x - a| < R$, where

$$R = \frac{1}{\limsup_{n \rightarrow \infty} |c_n|^{1/n}} \quad (\text{Cauchy-Hadamard}) \quad \text{or} \quad R = \lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right| \quad \text{when the limit exists.}$$

The series diverges for $|x - a| > R$; endpoints $|x - a| = R$ must be checked separately. Integration and differentiation do not change R .

2.2 Taylor and Maclaurin Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$$

Reading off derivatives: if $f(x) = \sum_{n=0}^{\infty} c_n x^n$, then

$$\boxed{f^{(k)}(0) = k! c_k}$$

To find $f^{(k)}(0)$: read off the coefficient of x^k in the series and multiply by $k!$. No need to differentiate the series k times explicitly.

2.3 Product and Quotient of Power Series

Cauchy product: if $f(x) = \sum_{n=0}^{\infty} b_n x^n$ and $g(x) = \sum_{n=0}^{\infty} c_n x^n$, then

$$f(x)g(x) = \sum_{n=0}^{\infty} a_n x^n, \quad a_n = \sum_{k=0}^n b_k c_{n-k}.$$

The radius of convergence of the product is at least $\min(R_f, R_g)$.

Example: $\frac{-\ln(1-x)}{1-x} = (\sum_{n=1}^{\infty} \frac{x^n}{n})(\sum_{n=0}^{\infty} x^n)$; the coefficient of x^n is $a_n = H_n = \sum_{k=1}^n \frac{1}{k}$ (harmonic number).

Quotient of power series: to expand $h(x) = P(x)/Q(x)$ near $x = 0$ (with $Q(0) \neq 0$):

1. Expand P and Q as power series to the required order.
2. Write $Q(x) = Q(0)(1 + r(x))$, $r(0) = 0$; use $1/(1+r) = 1 - r + r^2 - \dots$
3. Multiply $P(x) \cdot \frac{1}{Q(x)}$ term by term, then read off $h^{(k)}(0) = k! \cdot [\text{coeff of } x^k]$.

2.4 Common Maclaurin Series

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad R = \infty$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad R = \infty$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad R = \infty$$

$$\sinh x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}, \quad R = \infty \quad (\text{same as } \sin x \text{ without } (-1)^n)$$

$$\cosh x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}, \quad R = \infty \quad (\text{same as } \cos x \text{ without } (-1)^n)$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad |x| < 1$$

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}, \quad |x| \leq 1, x \neq -1$$

$$\arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}, \quad |x| \leq 1$$

$$(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n, \quad |x| < 1$$

Binomial coefficient reminder:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} \quad (\text{integer } n \geq k \geq 0, \text{ “}n \text{ choose } k\text{”})$$

$$\binom{\alpha}{k} = \frac{\alpha(\alpha-1)\cdots(\alpha-k+1)}{k!} \quad (\text{generalized, any } \alpha \in \mathbb{R})$$

Pascal's rule: $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$. Special values: $\binom{n}{0} = \binom{n}{n} = 1$, $\binom{n}{1} = n$.

2.5 Taylor's Theorem with Remainder

Theorem 2.1 (Lagrange Remainder). *If f has $n+1$ continuous derivatives on an interval containing a , then*

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + R_n(x), \quad R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1}$$

for some c strictly between a and x . Error bound:

$$|R_n(x)| \leq \frac{M_{n+1}}{(n+1)!} |x-a|^{n+1}, \quad M_{n+1} = \sup_{t \text{ between } a, x} |f^{(n+1)}(t)|.$$

2.6 Applications: Limits and Integrals via Taylor

Computing limits at 0: expand numerator and denominator as power series, cancel common x^k factors, read off the limit from the leading terms.

Computing integrals: substitute a known series for the integrand and integrate term by term (valid inside the radius of convergence). Use the Leibniz error bound (Theorem 1.2) to determine how many terms are needed.

3 Uniform Convergence

3.1 Sequences of Functions

Definition 3.1. Let $f_n, f : D \rightarrow \mathbb{R}$.

- **Pointwise:** $f_n \rightarrow f$ if for each $x \in D$, $f_n(x) \rightarrow f(x)$.
- **Uniform** ($f_n \Rightarrow f$): $\sup_{x \in D} |f_n(x) - f(x)| \rightarrow 0$.

Uniform $\Rightarrow$ pointwise, but pointwise $\not\Rightarrow$ uniform in general.

Standard method: compute $M_n = \sup_{x \in D} |f_n(x) - f(x)|$ (find the max by differentiation). If $M_n \rightarrow 0$: uniform. If $M_n \not\rightarrow 0$: not uniform.

Negation: $f_n \not\Rightarrow f$ iff $\exists \varepsilon > 0$ and a sequence $x_n \in D$ with $|f_n(x_n) - f(x_n)| \geq \varepsilon$. Typical choice: $x_n = \arg \max_x |f_n(x) - f(x)|$.

Example 3.2 ($f_n(x) = x^n$ on $[0, 1]$). Pointwise limit: $f(x) = 0$. But $\sup_{x \in [0, 1]} |x^n| = 1 \not\rightarrow 0$. Not uniform. On $[0, r]$ for any $r < 1$: $\sup = r^n \rightarrow 0$. Uniform.

3.2 Properties Preserved by Uniform Convergence

Theorem 3.3 (Continuity). *If each f_n is continuous on D and $f_n \Rightarrow f$, then f is continuous on D .*

Theorem 3.4 (Term-by-term Integration). *If $f_n \Rightarrow f$ on $[a, b]$, then $\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b f(x) dx$.*

Theorem 3.5 (Term-by-term Differentiation). *If $f_n \rightarrow f$ pointwise on $[a, b]$, each f_n is C^1 , and $f'_n \Rightarrow g$ uniformly, then f is differentiable and $f' = g$.*

Theorem 3.6 (Sequential Substitution). *If $f_n \Rightarrow f$ uniformly on $[a, b]$ and $x_n \rightarrow x \in [a, b]$, then $f_n(x_n) \rightarrow f(x)$.*

Remark (Swapping $\sum$ and improper integrals). For $\int_0^\infty \sum u_n(x) dx$: if each $u_n \geq 0$ and $\sum \int_0^\infty u_n < \infty$, the swap is valid. Otherwise: (1) justify the swap on each compact $[a, b]$ via uniform convergence; (2) show the endpoint contributions (near 0 and near ∞) vanish as $a \rightarrow 0^+$, $b \rightarrow \infty$.

3.3 Series of Functions

A series $\sum_{n=1}^\infty u_n(x)$ **converges uniformly** on D if its partial sums converge uniformly on D .

Domain of convergence: treat x as a parameter; apply standard numerical-series tests to $\sum u_n(x)$ for each fixed x .

Theorem 3.7 (Weierstrass M -Test). *If $|u_n(x)| \leq M_n$ for all $x \in D$ and $\sum M_n < \infty$, then $\sum u_n(x)$ converges uniformly and absolutely on D .*

Remark. The M -test is sufficient but *not necessary*: a series can converge uniformly even when $\sum M_n = \infty$. When the M -test fails, try the uniform Dirichlet or Abel tests.

Example (M -test fails, Dirichlet works): $\sum_{n=1}^\infty \frac{(-1)^n}{n}$ write as $\sum a_n b_n$ with $a_n = (-1)^n$, $b_n = 1/n$.

- Partial sums of $\sum (-1)^n$: $|A_N| \leq 1$ for all N (bounded).
- $b_n = 1/n \searrow 0$ monotonically (no x -dependence, so trivially uniform).

By the uniform Dirichlet test, $\sum (-1)^n/n$ converges uniformly on $\mathbb{R}$, yet $\sum 1/n = \infty$ so the M -test cannot be applied.

Theorem 3.8 (Dirichlet — Uniform Version). *$\sum a_n(x) b_n(x)$ converges uniformly on D if the partial sums $A_N(x)$ are uniformly bounded and $b_n(x) \searrow 0$ monotonically and uniformly on D .*

Theorem 3.9 (Abel — Uniform Version). *$\sum a_n(x) b_n(x)$ converges uniformly on D if $\sum a_n(x)$ converges uniformly and $b_n(x)$ is monotone and uniformly bounded on D .*

Power series: $\sum c_n x^n$ with radius R converges uniformly on $[-r, r]$ for any $r < R$, but generally not on $(-R, R)$.

Properties of uniformly convergent series: the sum is continuous if each u_n is; term-by-term integration is valid; term-by-term differentiation requires $\sum u'_n$ to converge uniformly.

4 Inner Product Spaces & L^2 Norm

4.1 Inner Product on $L^2[-L, L]$

$$\langle f, g \rangle = \int_{-L}^L f(x) \overline{g(x)} dx, \quad \|f\| = \sqrt{\langle f, f \rangle} = \sqrt{\int_{-L}^L |f(x)|^2 dx}$$

4.2 Orthogonality of the Trigonometric Basis

The functions $\{1, \cos(n\pi x/L), \sin(n\pi x/L)\}_{n \geq 1}$ are orthogonal on $[-L, L]$:

$$\int_{-L}^L \cos \frac{n\pi x}{L} \cos \frac{m\pi x}{L} dx = \begin{cases} 2L & n = m = 0 \\ L & n = m \neq 0 \\ 0 & n \neq m \end{cases} \quad \int_{-L}^L \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L} dx = \begin{cases} L & n = m \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\int_{-L}^L \sin \frac{n\pi x}{L} \cos \frac{m\pi x}{L} dx = 0 \quad \text{for all } n, m$$

4.3 Bessel's Inequality and Parseval's Identity

Bessel's Inequality	Parseval's Identity
$\frac{a_0^2}{2} + \sum_{n=1}^N (a_n^2 + b_n^2) \leq \frac{1}{L} \ f\ ^2$ for every finite N. Holds for any $f \in L^2[-L, L]$, even if the Fourier series does not converge to f pointwise.	$\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) = \frac{1}{L} \ f\ ^2$ (holds for all $f \in L^2[-L, L]$). Parseval = Bessel with equality. Equivalently: $\ S_N - f\ \rightarrow 0$ (convergence in norm).

Convergence implications (on a bounded interval): uniform $\Rightarrow$ pointwise; uniform $\Rightarrow L^2$ norm; pointwise $\not\Rightarrow L^2$; $L^2 \not\Rightarrow$ pointwise.

5 Fourier Series

5.1 Definition and Coefficients

For f periodic with period $2L$:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx, \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx$$

The integral may be taken over any full period, not only $[-L, L]$.

5.2 Pointwise Convergence: Dirichlet's Theorem

If f is piecewise smooth and periodic, the Fourier series $S(x)$ converges to:

- $f(x)$ at points of continuity,
- $\frac{f(x^+) + f(x^-)}{2}$ at jump discontinuities.

Evaluating $S(x_0)$ outside $[-L, L]$: The Fourier series represents the *periodic extension* of f (period $2L$). For $x_0 \notin [-L, L]$: reduce x_0 modulo $2L$ to $x'_0 \in [-L, L]$, then apply Dirichlet's theorem at x'_0 .

Example: f on $[-\pi, \pi]$, find $S(3\pi)$. Since $3\pi \equiv \pi \pmod{2\pi}$, and π is a jump of the periodic extension: $S(3\pi) = S(\pi) = \frac{f(\pi^-) + f(-\pi^+)}{2}$.

5.3 Even, Odd, and Half-Period Extensions

Even f : $b_n = 0$; $a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx$

Odd f : $a_n = 0$; $b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$

Given f on $[0, L]$, extend to $[-L, L]$ (evenly or oddly) then periodically:

Cosine series (even extension):

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L}, \quad a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx$$

Sine series (odd extension):

$$f(x) \sim \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}, \quad b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$$

Both converge to $f(x)$ on $(0, L)$ at continuity points. At endpoints: cosine series converges to the actual endpoint values; sine series is forced to 0 (odd extension).

5.4 Shift Property

If f has period 2π with coefficients a_n, b_n , and $g(x) = f(x + h)$, then:

$$A_n = a_n \cos(nh) + b_n \sin(nh), \quad B_n = -a_n \sin(nh) + b_n \cos(nh)$$

Special case $h = \pi$: $A_n = (-1)^n a_n$, $B_n = (-1)^n b_n$, $A_0 = a_0$.

5.5 Parseval's Identity and Computing Numerical Series

$$\frac{1}{L} \int_{-L}^L |f(x)|^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

Method 1 — substitute $x = x_0$: at a continuity point $S(x_0) = f(x_0)$; at a jump $S(x_0) = \frac{f(x_0^+) + f(x_0^-)}{2}$. Useful values: $x = 0$ (kills sin terms), $x = \pi$ ($\cos(n\pi) = (-1)^n$), $x = \pi/2$.

Method 2 — Parseval: compute the left side as a definite integral; equate to the right side to extract the desired sum.

Reference table:

Series	Value
$\sum_{n=1}^{\infty} \frac{1}{n^2}$	$\frac{\pi^2}{6}$
$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$	$\frac{\pi^2}{12}$
$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}$	$\frac{\pi^2}{8}$
$\sum_{n=1}^{\infty} \frac{1}{n^4}$	$\frac{\pi^4}{90}$
$\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$	$\frac{\pi}{4}$
$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^3}$	$\frac{\pi^3}{32}$

Remark. $\sum 1/(2n+1)^2 = \pi^2/8$ follows from $\sum 1/n^2 = \pi^2/6$ by separating even and odd terms: $\sum_{n=1}^{\infty} \frac{1}{(2n)^2} = \frac{1}{4} \cdot \frac{\pi^2}{6}$, so $\sum_{\text{odd}} = \frac{\pi^2}{6} - \frac{\pi^2}{24} = \frac{\pi^2}{8}$.

5.6 Complex Fourier Series

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/L}, \quad c_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-in\pi x/L} dx$$

Relationship to real coefficients ($n \geq 1$):

$$c_0 = \frac{a_0}{2}, \quad c_n = \frac{a_n - ib_n}{2}, \quad c_{-n} = \frac{a_n + ib_n}{2}, \quad a_n = c_n + c_{-n}, \quad b_n = i(c_n - c_{-n})$$

Quick route: if you already have the real series, read off c_n without recomputing integrals.

$$\text{Parseval for complex series: } \frac{1}{2L} \int_{-L}^L |f(x)|^2 dx = \sum_{n=-\infty}^{\infty} |c_n|^2$$

5.7 Summing a Real Fourier Series in Closed Form

Geometric series technique: To find a closed form for $\sum_{n=1}^{\infty} r^n \sin(nx)$ or $\sum r^n \cos(nx)$ ($0 < r < 1$):

1. Write as Im or Re of $\sum_{n=1}^{\infty} (re^{ix})^n$.

2. Sum as a geometric series: $\sum_{n=1}^{\infty} z^n = \frac{z}{1-z}, \quad |z| < 1$.

3. Multiply numerator and denominator by $\overline{(1-z)}$; separate real and imaginary parts.

Example:

$$\sum_{n=1}^{\infty} \frac{\sin(nx)}{3^n} = \operatorname{Im} \left(\frac{e^{ix}/3}{1 - e^{ix}/3} \right) = \operatorname{Im} \left(\frac{3e^{ix}}{3 - e^{ix}} \right) = \frac{3 \sin x}{10 - 6 \cos x}$$

(denominator $|3 - e^{ix}|^2 = 10 - 6 \cos x$; imaginary part of numerator $= 3 \sin x$).

Infinite differentiability via the M -test on derivative series: $\sum u_n$ is C^∞ on D if for every $k \geq 0$ the series $\sum u_n^{(k)}$ passes the M -test (i.e. $\sum_n \|u_n^{(k)}\|_\infty < \infty$). **Example:** $|n^k \sin(nx)/3^n| \leq n^k/3^n$, and $\sum n^k/3^n < \infty$ for all k (geometric dominates polynomial), so $\sum \sin(nx)/3^n \in C^\infty(\mathbb{R})$.

5.8 Hyperbolic and Complex Functions (for Fourier)

Definitions and key identity:

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}, \quad e^x = \cosh x + \sinh x$$

Properties: $\cosh^2 x - \sinh^2 x = 1$; $(\sinh x)' = \cosh x$; $(\cosh x)' = \sinh x$; $\sinh(-x) = -\sinh x$ (odd); $\cosh(-x) = \cosh x$ (even).

Euler's formula: $e^{i\theta} = \cos \theta + i \sin \theta$

All four functions via e :

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} \quad \cosh x = \frac{e^x + e^{-x}}{2} \quad \sinh x = \frac{e^x - e^{-x}}{2}$$

Cross-relations (multiplying argument by i swaps trig $\leftrightarrow$ hyperbolic):

$$\cos(ix) = \cosh x, \quad \sin(ix) = i \sinh x, \quad \cosh(ix) = \cos x, \quad \sinh(ix) = i \sin x$$

Complex argument formulas:

$$\sinh(a+ib) = \sinh(a) \cos(b) + i \cosh(a) \sin(b), \quad \cosh(a+ib) = \cosh(a) \cos(b) + i \sinh(a) \sin(b)$$

Key simplification for integer n (constant in complex Fourier):

$$\sinh(a - in\pi) = (-1)^n \sinh(a), \quad \cosh(a - in\pi) = (-1)^n \cosh(a)$$

because $\cos(n\pi) = (-1)^n$ and $\sin(n\pi) = 0$.

5.9 Technique: Complex Fourier of e^{ax} on $[-L, L]$

$$c_n = \frac{1}{2L} \int_{-L}^L e^{ax} e^{-in\pi x/L} dx = \frac{(-1)^n \sinh(aL)}{L(a - in\pi/L)} = \frac{(-1)^n \sinh(aL) (a + in\pi/L)}{L(a^2 + n^2\pi^2/L^2)}$$

Reading off the real coefficients:

$$a_n = \frac{2(-1)^n \sinh(aL) \cdot a}{L(a^2 + n^2\pi^2/L^2)}, \quad b_n = \frac{2(-1)^n \sinh(aL) \cdot n\pi/L}{L(a^2 + n^2\pi^2/L^2)}$$

Even/odd shortcut: write $e^{ax} = \cosh(ax) + \sinh(ax)$. Since $\cosh$ is even it contributes only to a_n ; since $\sinh$ is odd it contributes only to b_n . This halves the computation.

6 Fourier Transform

6.1 Definition and Inverse (ω convention)

$$\hat{f}(\omega) = \mathcal{F}\{f\}(\omega) = \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx, \quad f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i\omega x} d\omega$$

Fourier Integral Representation (real form):

$$f(x) = \int_0^{\infty} [A(\omega) \cos(\omega x) + B(\omega) \sin(\omega x)] d\omega, \quad A(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(t) \cos(\omega t) dt, \quad B(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(t) \sin(\omega t) dt$$

Pointwise convergence of the IFT: if $f \in L^1(\mathbb{R})$ and f is piecewise smooth, the IFT formula holds at every point of continuity of f .

Technique — extracting definite integrals from the IFT: substituting $x = x_0$ gives

$$\int_{-\infty}^{\infty} \hat{f}(\omega) e^{i\omega x_0} d\omega = 2\pi f(x_0).$$

For even f (so $\hat{f}$ is real): $\int_0^{\infty} \hat{f}(\omega) \cos(\omega x_0) d\omega = \pi f(x_0)$.

6.2 Key Properties

Property	Formula
Linearity	$\mathcal{F}\{af + bg\} = a\hat{f} + b\hat{g}$
Time shift	$\mathcal{F}\{f(x - a)\}(\omega) = e^{-i\omega a} \hat{f}(\omega)$
Freq. shift	$\mathcal{F}\{e^{iax} f(x)\}(\omega) = \hat{f}(\omega - a)$
Scaling	$\mathcal{F}\{f(ax)\}(\omega) = \frac{1}{ a } \hat{f}\left(\frac{\omega}{a}\right)$
Derivative	$\mathcal{F}\{f'(x)\}(\omega) = i\omega \hat{f}(\omega)$
Mult. by x	$\mathcal{F}\{xf(x)\}(\omega) = i \frac{d\hat{f}}{d\omega}(\omega)$
Convolution	$\mathcal{F}\{f * g\} = \hat{f} \cdot \hat{g}$
Duality	$\mathcal{F}\{\hat{f}(x)\}(\omega) = 2\pi f(-\omega)$
Cos modulation	$\mathcal{F}\{f(x) \cos(ax)\} = \frac{\hat{f}(\omega - a) + \hat{f}(\omega + a)}{2}$
Sin modulation	$\mathcal{F}\{f(x) \sin(ax)\} = \frac{\hat{f}(\omega - a) - \hat{f}(\omega + a)}{2i}$
Even f	$\hat{f}(\omega)$ is real
Odd f	$\hat{f}(\omega)$ is purely imaginary
Riemann–Lebesgue	If $f \in L^1(\mathbb{R})$: $\hat{f}$ is continuous on $\mathbb{R}$ and $\hat{f}(\omega) \rightarrow 0$ as $ \omega \rightarrow \infty$
Half-line (even f)	$\int_0^{\infty} f(x) \cos(\omega x) dx = \frac{1}{2} \hat{f}(\omega)$ (since $\hat{f}(\omega) = 2 \int_0^{\infty} f(x) \cos(\omega x) dx$ when f is even)

6.3 Common Transform Pairs

$f(x)$	$\hat{f}(\omega)$	Notes
$e^{-ax} \mathbf{1}_{x \geq 0}$	$\frac{1}{a + i\omega}$	$a > 0$
$e^{-a x }$	$\frac{2a}{a^2 + \omega^2}$	$a > 0$
$\frac{1}{x^2 + a^2}$	$\frac{\pi}{a} e^{-a \omega }$	$a > 0$
e^{-ax^2}	$\sqrt{\frac{\pi}{a}} e^{-\omega^2/(4a)}$	$a > 0$, Gaussian $\rightarrow$ Gaussian
$\mathbf{1}_{ x \leq L}$	$\frac{2 \sin(L\omega)}{\omega}$	rect function

6.4 Parseval / Plancherel Theorem

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\hat{f}(\omega)|^2 d\omega$$

7 Laplace Transform

7.1 Definition

$$\mathcal{L}\{f\}(s) = F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

7.2 Common Transforms and Properties

$f(t)$	$F(s)$
1	$\frac{1}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s - a}$
$\sin(bt)$	$\frac{b}{s^2 + b^2}$
$\cos(bt)$	$\frac{s}{s^2 + b^2}$
$\sinh(bt)$	$\frac{b}{s^2 - b^2}, \quad s > b $
$\cosh(bt)$	$\frac{s}{s^2 - b^2}, \quad s > b $
$e^{at} f(t)$	$F(s - a)$ (frequency shift)
$f'(t)$	$sF(s) - f(0)$
$f''(t)$	$s^2 F(s) - sf(0) - f'(0)$
$(f * g)(t)$	$F(s) \cdot G(s)$ (convolution)